SMARTX: Porewater Methane

September 2025 Samples

2026-01-09

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##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

```
## Run Information: Brenden D. Blakley
```

```
#set the run date & user name
sample_year <- "2025"
sample_month <- "September"
user <- "Brenden Blakley"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

GHG_files <- c("Raw Data/20251125_GHGdata_SMARTX_Sept.csv","Raw Data/20251124_GHGdata_SMARTX_Sept.csv",


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/SMARTX_ShimadzuGC_QAQC_2025.csv"


# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_SMARTX_Porewater_CH4_202509.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)

volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrized
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrized (should be equal to volume_g

#record any notes about the run or anything other info here:
run_notes <- "Some Blanks Had Check Standard instead"

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

```
## Some Blanks Had Check Standard instead
```

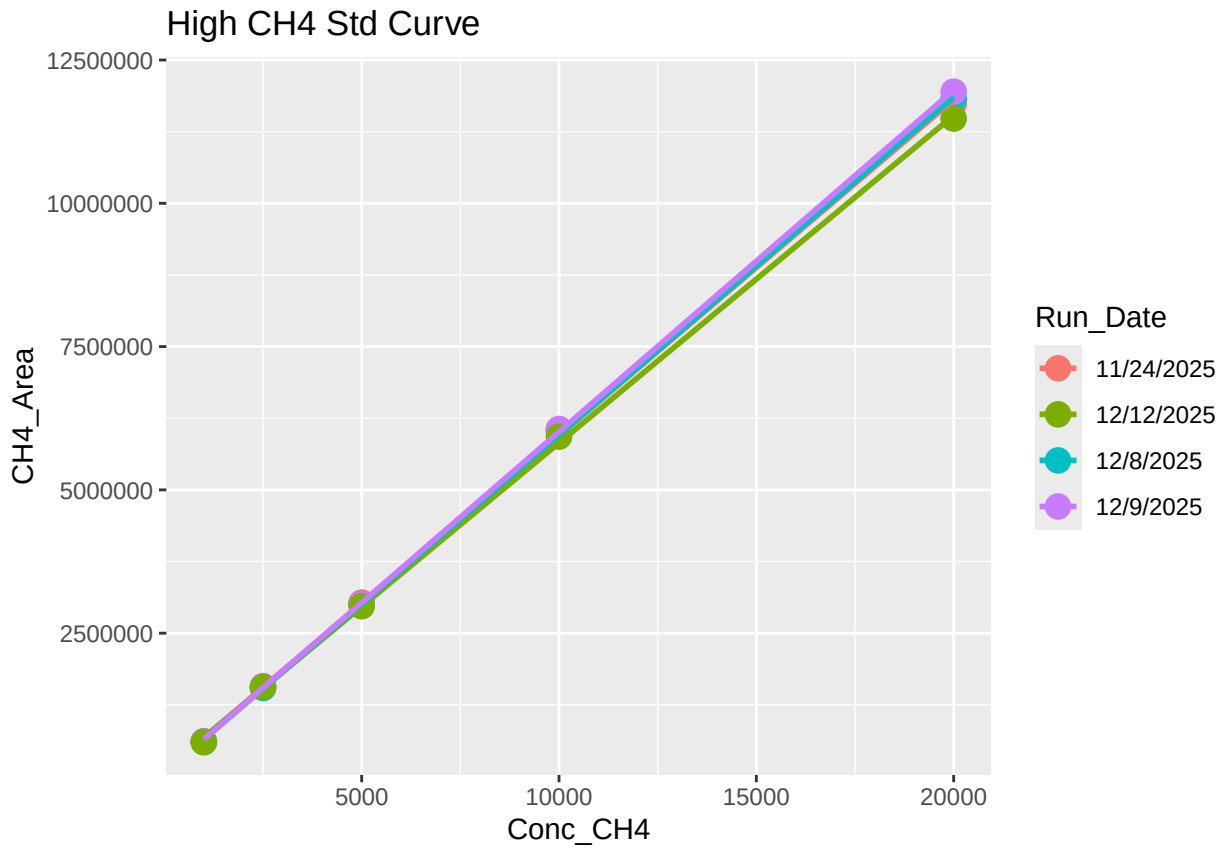
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

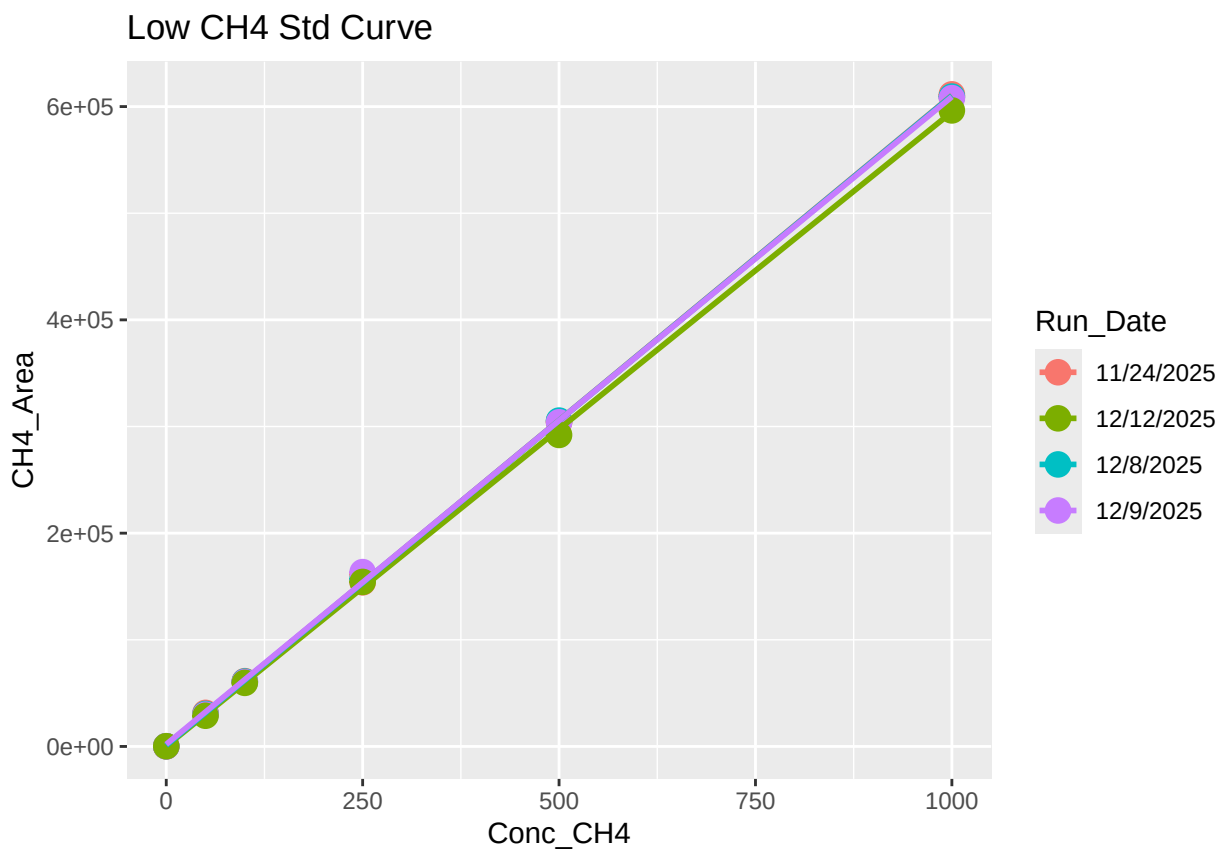
1 Read in data files

2 Assess standard curves

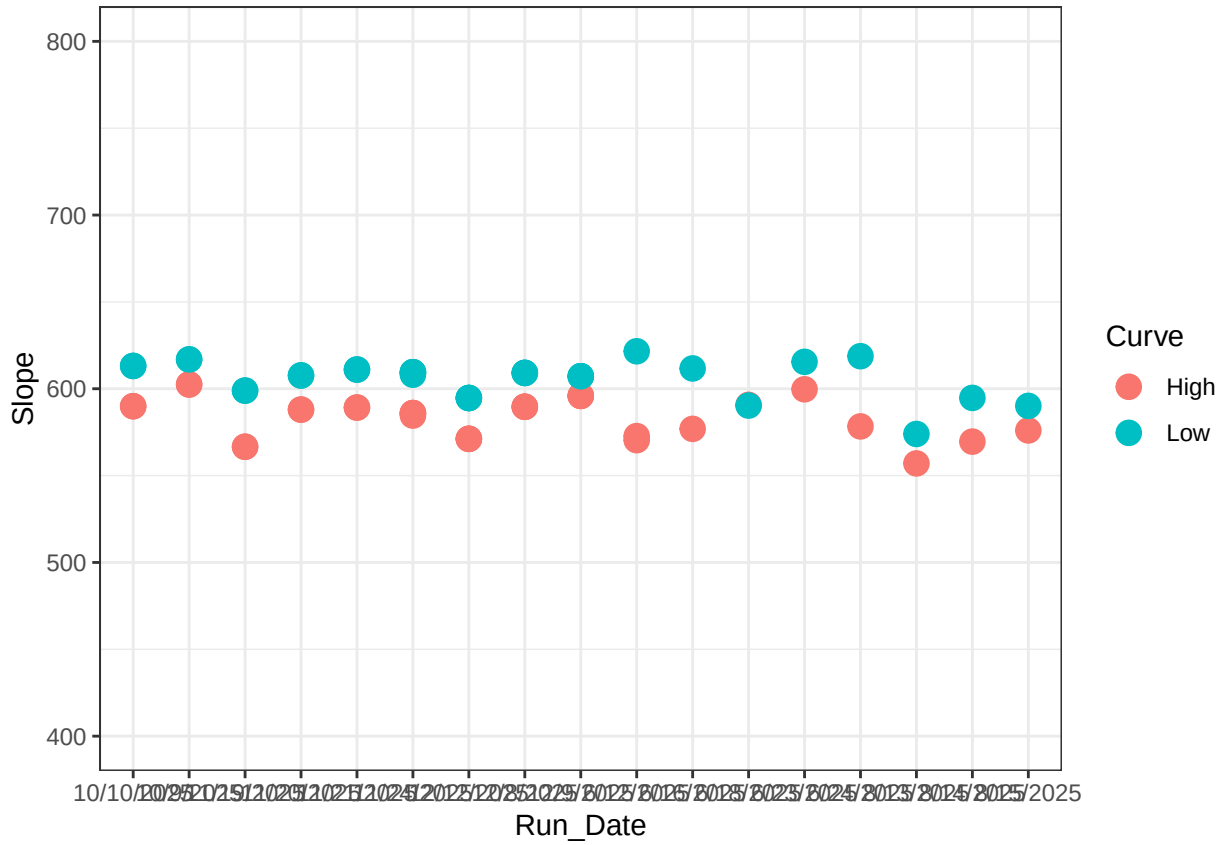
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
##   X Curve        R2    Slope    Intercept  Run_Date
## 1 1  High 0.9986295 576.8990  173986.2174 6/16/2025
## 2 2   Low 0.9990260 611.6961   -3791.2684 6/16/2025
## 3 3  High 0.9999101 570.3925  138747.3478 6/12/2025
## 4 4   Low 0.9979098 621.5734    3224.2535 6/12/2025
## 5 5  High 0.9989882 590.7374 -109278.7391 6/18/2025
## 6 6   Low 0.9990995 590.3138     954.7682 6/18/2025
```



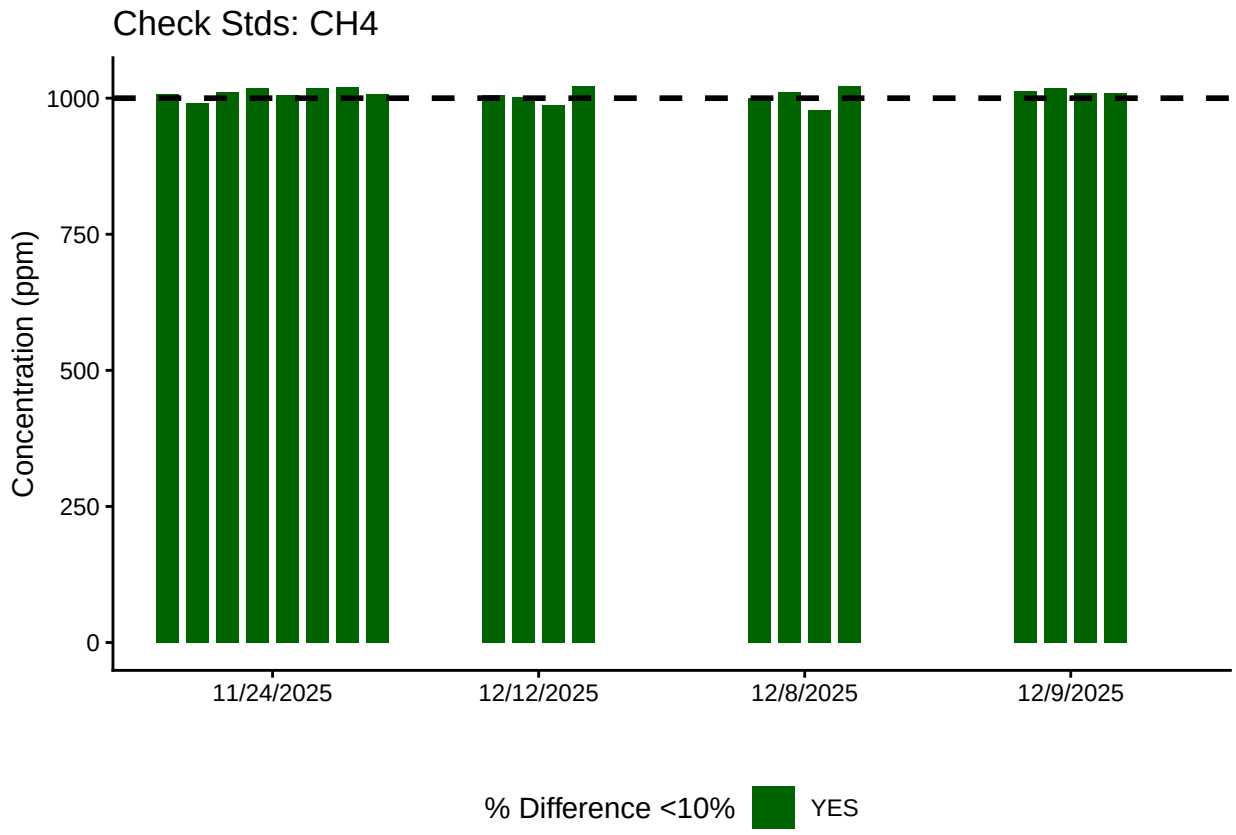

```
## [1] "High_R2 is Good-11/24/2025" "High_R2 is Good-12/12/2025"
## [3] "High_R2 is Good-12/8/2025"  "High_R2 is Good-12/9/2025"
```

```
## [1] "Low_R2 is Good-11/24/2025" "Low_R2 is Good-12/12/2025"
## [3] "Low_R2 is Good-12/8/2025"  "Low_R2 is Good-12/9/2025"
```

2.1 Now calculate the CH4 in ppm

3 Aanlayze the Check Standards

```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

```
## Assess Blanks
```

```
## >60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED
```

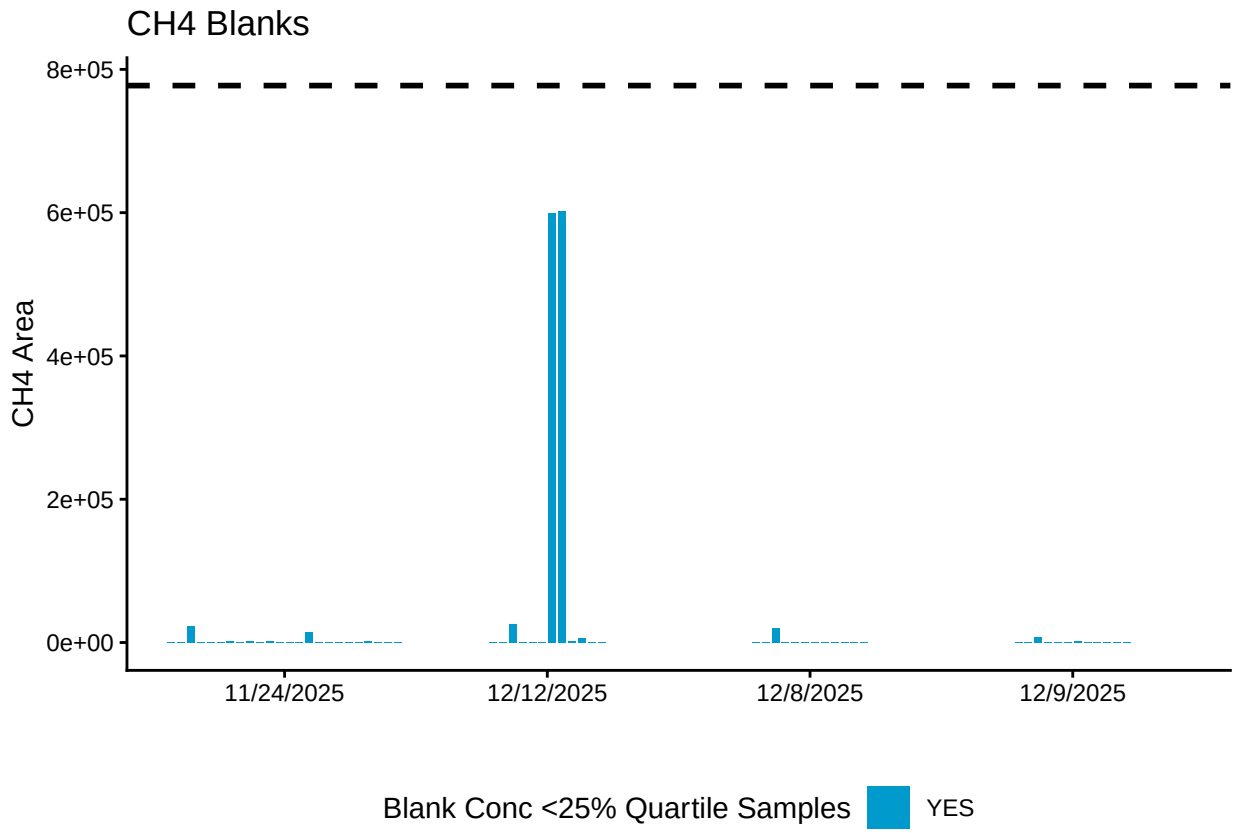
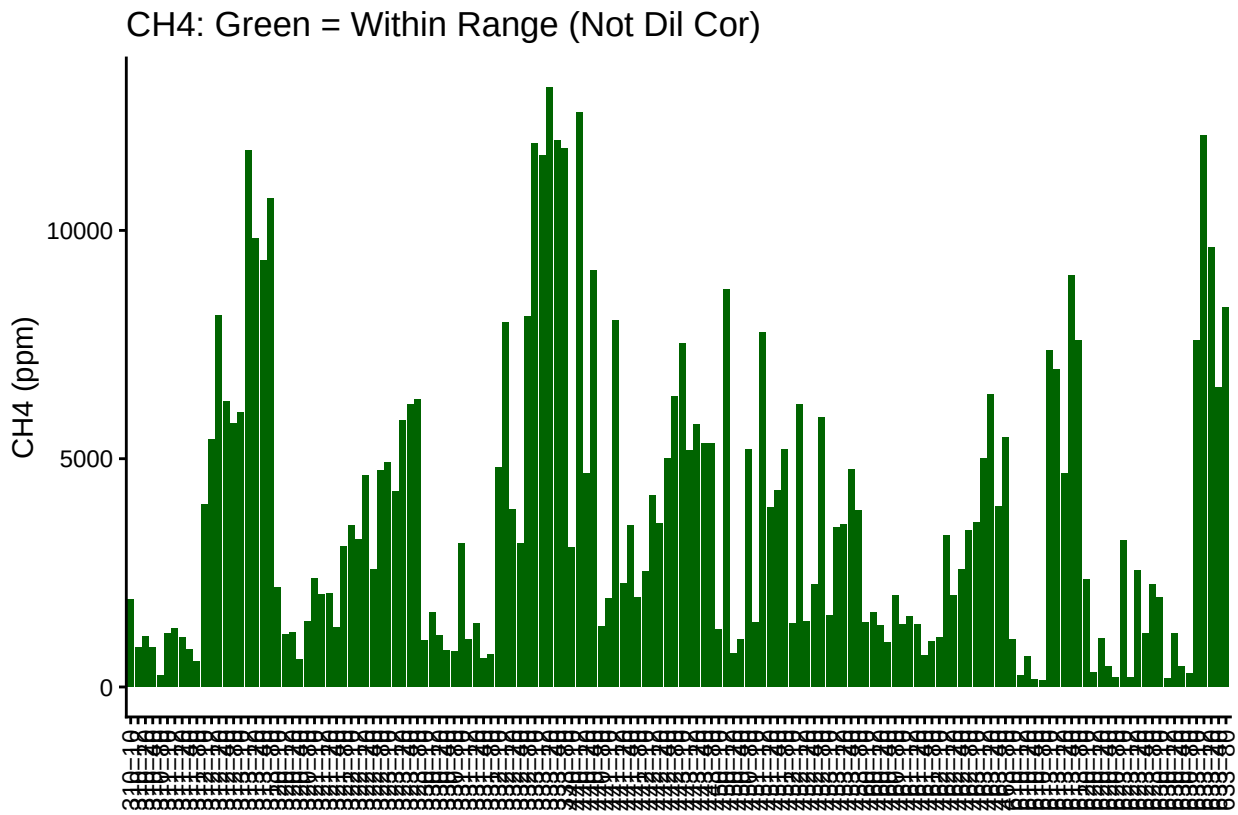
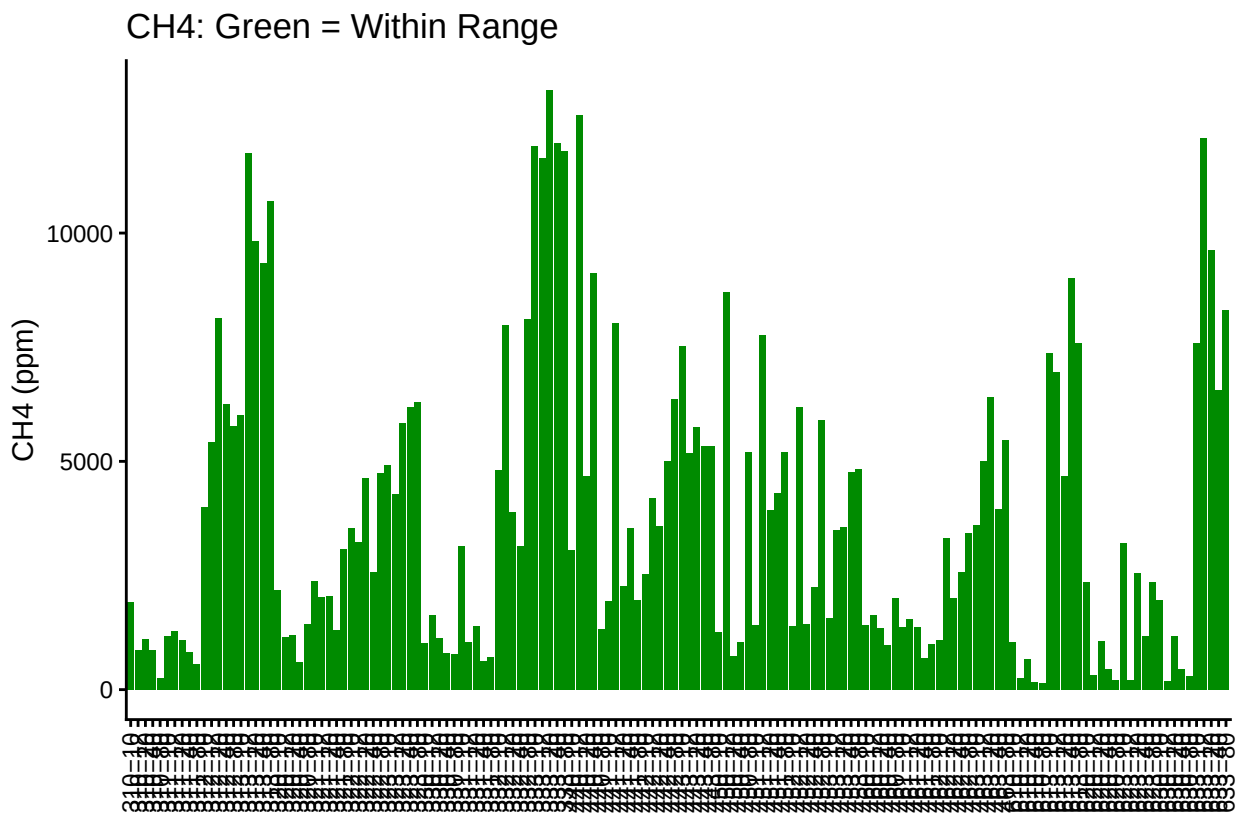


Table 1: Mean Area of Blanks

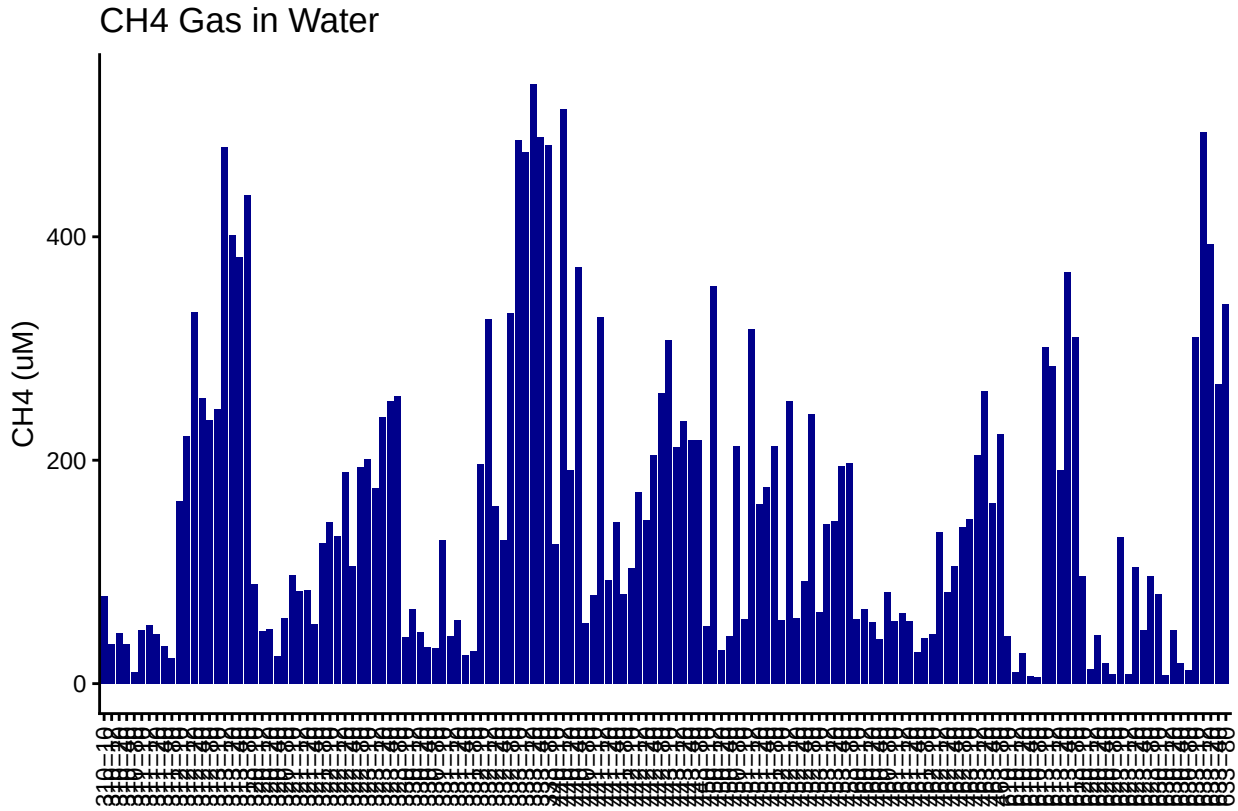
Test	Blank_Mean_Area
CH4	22216.73

5 Dilution correct samples and Check Range





6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

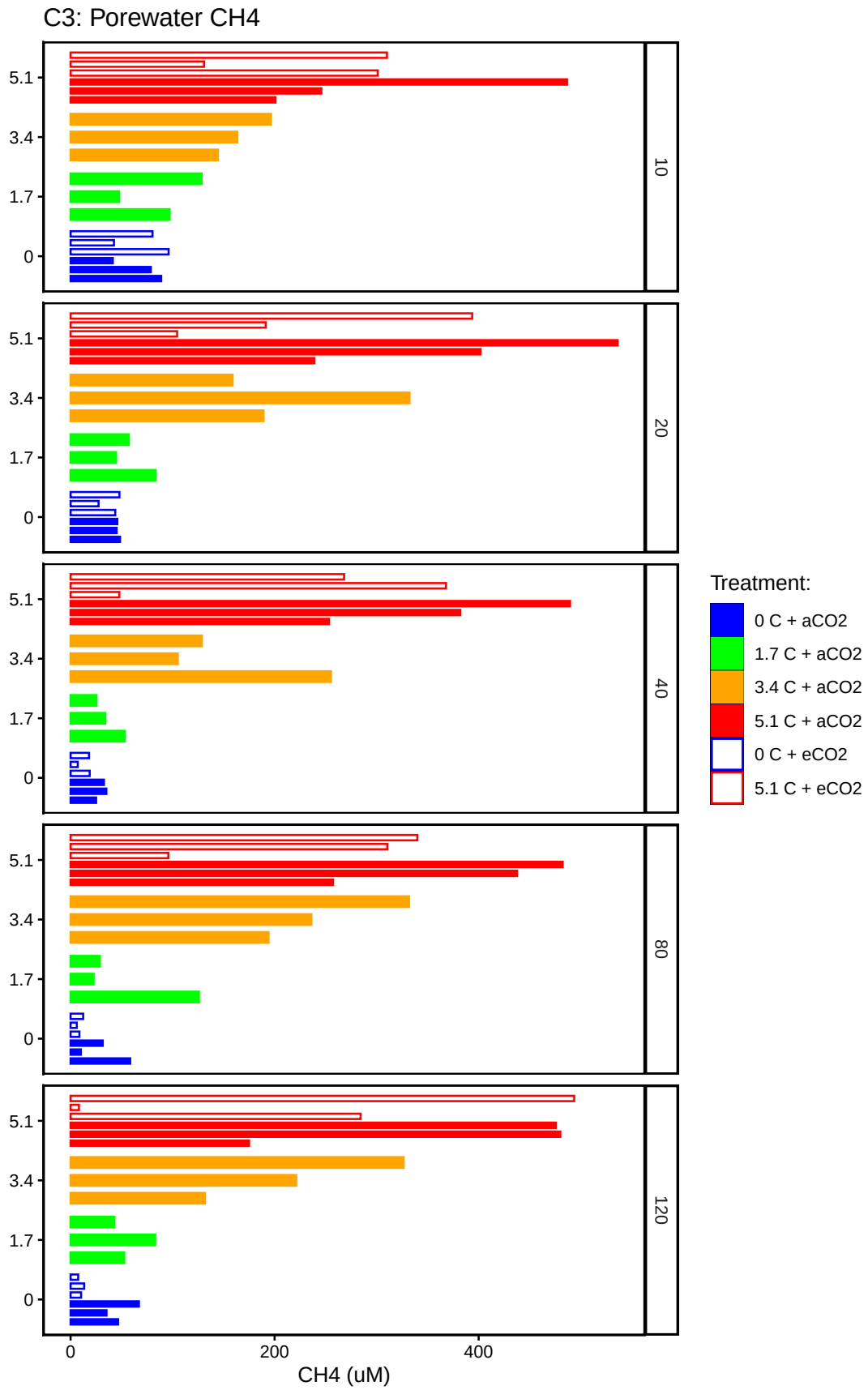
```
## Merge Samples with Metadata
```

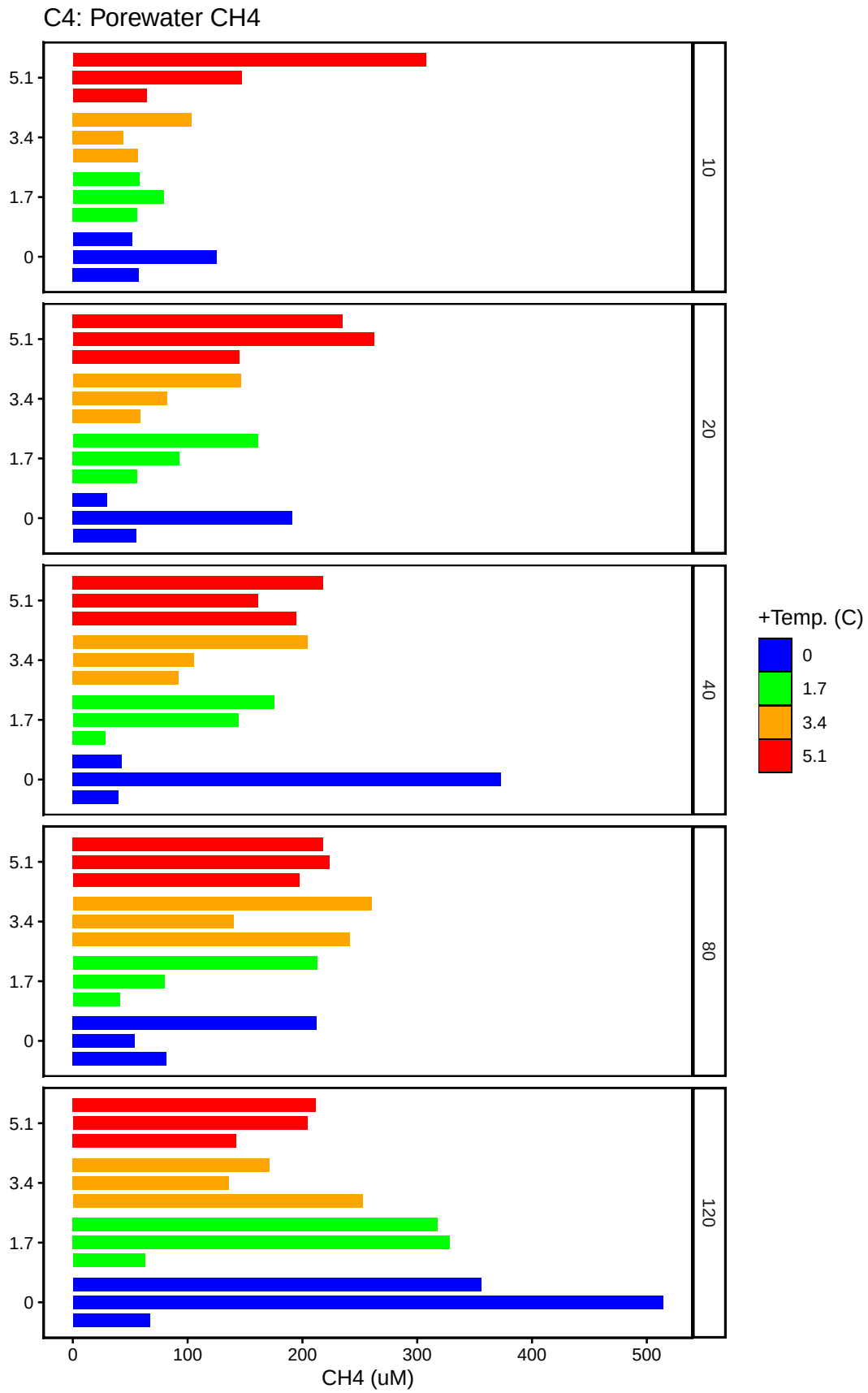
```
## All sample IDs are present in data
```

7.1 Organize Data

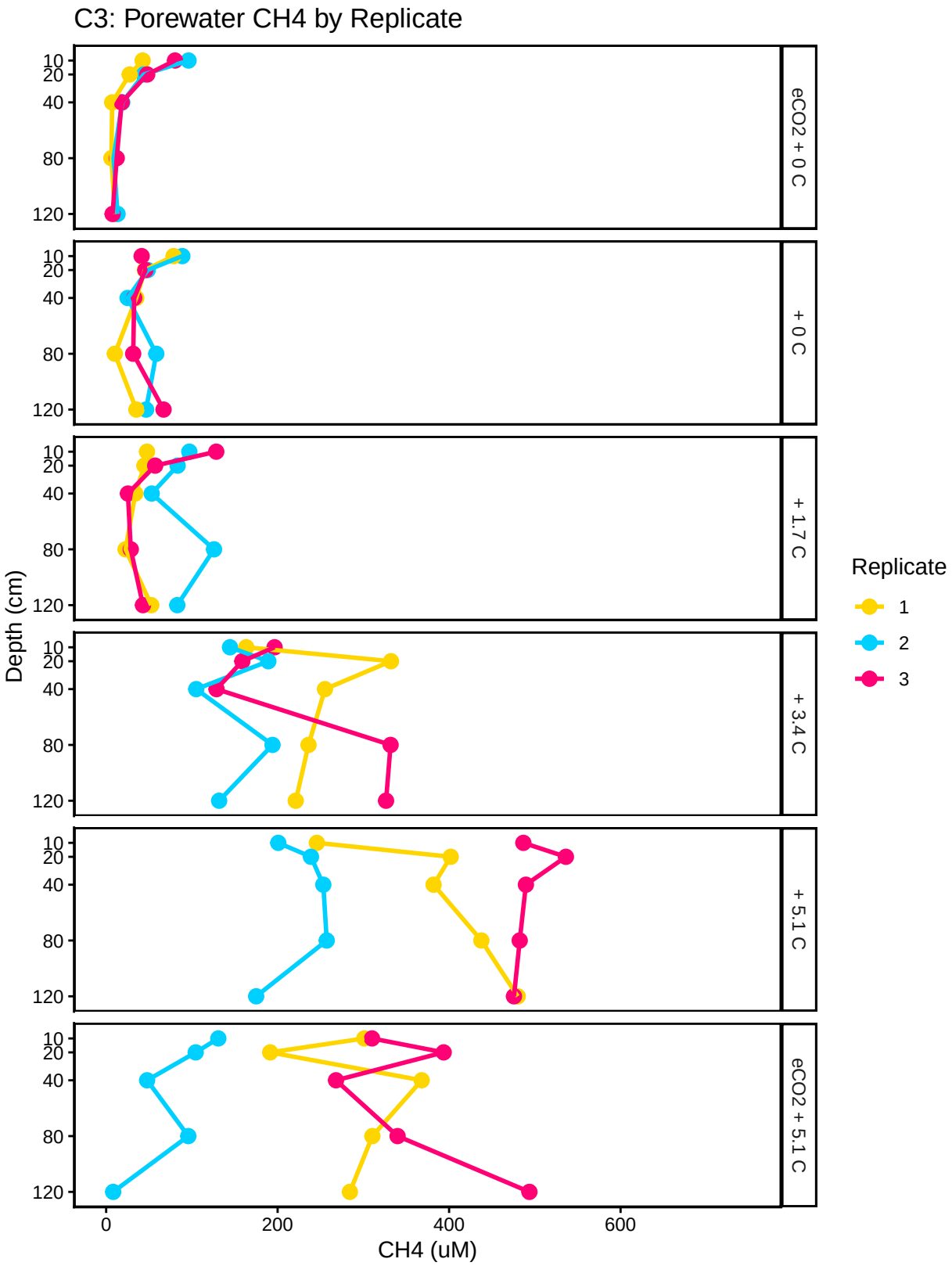
7.2 Visualize Data

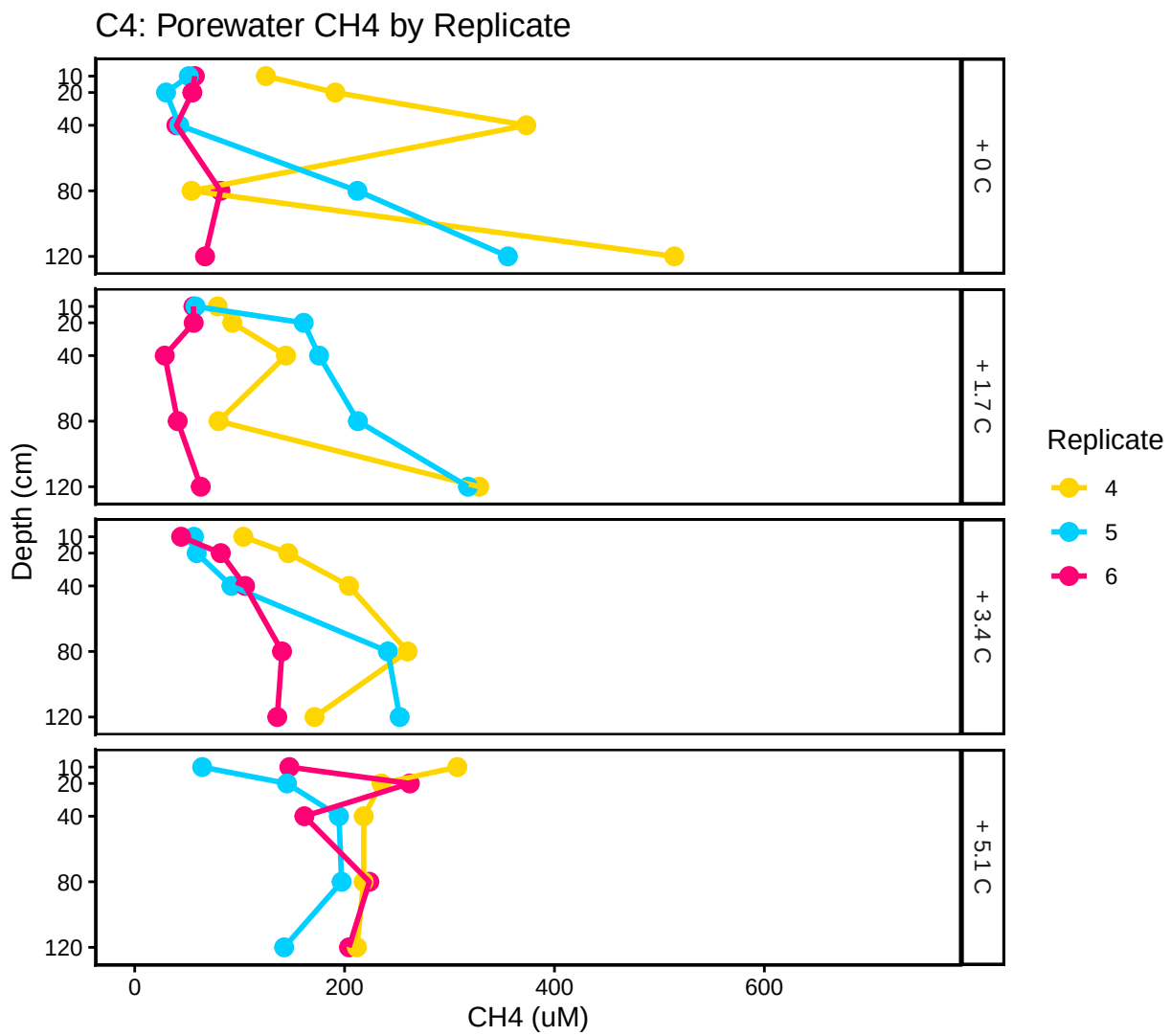
```
## Visualize Data
```


C3: Porewater CH4
5.1
3.4
1.7
0
10
20
40
80
120
0
200
400
CH4 (uM)
Treatment:
0 C + aCO2
1.7 C + aCO2
3.4 C + aCO2
5.1 C + aCO2
0 C + eCO2
5.1 C + eCO2


C4: Porewater CH4
5.1
3.4
1.7
0
10
20
40
80
120
0
100
200
300
400
500
CH4 (uM)
+Temp. (C)
0
1.7
3.4
5.1

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```


C3: Porewater CH4 by Replicate
Depth (cm)
10
20
40
80
120
eCO2 + 0 C
+ 0 C
+ 1.7 C
+ 3.4 C
+ 5.1 C
eCO2 + 5.1 C
0
200
400
600
CH4 (uM)
Replicate
1
2
3



##Export Data

#end